

Antoine Lucas

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R / Shiny Engineer · Scientific Software & Data Products

Pharma · Cosmetics · Biotech · Paris, France

Research Engineer & Data Scientist with 5+ years building production-grade software and data products in pharma, cosmetics, and biotech. R / Shiny delivery, reproducible analysis, stakeholder-facing architecture, and technical ownership from discovery through deployment. Background spans statistical modelling, scientific computing, internal platform tooling, and ML systems, with a strong interest toward maintainable delivery: documentation, validation, CI/CD, and handover for teams that need to keep using the product long after the first release.

Selected Experience

Data Scientist & ML Engineer

Sep 2025 – present

Chanel Parfums Beauté R&D · via Astek / IT&M Stats

- Delivered R Shiny applications covering sensory scoring, clinical data visualisation, interactive experiment planning for R&D scientists, enhanced data lifecycle management for researchers (20+ app catalog)
- Responsible for data science platform tooling: Azure ML, Posit Connect, Databricks (deployment, access governance, and lifecycle management)
- Engineered reproducible MLOps and delivery infrastructure: Docker, GitHub Actions, drift monitoring, and qualification-minded workflows in a Good Practices (GxP)-adjacent environment
- Designed and deployed computer vision models (object segmentation) for cosmetics formulation and regulation R&D

Data Scientist — Internal & Consulting

May – Sep 2025

Astek / IT&M Stats

- Led internal upskilling training on bayesian statistics with R for internal consultants
- Developed reusable internal R packages and Quarto automated reporting templates, cutting analyst turnaround time by 25% across consulting engagements
- Scoped and prototyped ML solutions for pre-sales client engagements in pharma and agri-food verticals
- Fine-tuned Microsoft Phi-3-mini-4k-instruct (3.8B) using LoRA (PEFT) on the climatecheck dataset for claim verification — full pipeline from data preparation to evaluation

Data Scientist — Multi-omics

Dec 2024 – May 2025

Abolis · Microbiome Studio · via Astek / IT&M Stats

- Context: Fast-paced 8-person startup contributing to a software-as-a-service (SaaS) product (FastAPI Python backend, Vue.js front-end), with full individual autonomy
- Delivered end-to-end pipelines in R and Python for tabular and large-scale omics data (genomics, transcriptomics, metabolomics)
- Integrated analysis outputs into the production SaaS platform via representational state transfer (REST) API

Data Scientist

Jul 2023 – Nov 2024

Sanofi R&D · Manufacturing Chain · via Astek / IT&M Stats

- Owned end-to-end delivery as sole R Shiny developer and Scrum Master, managing backlog, sprint ceremonies, and stakeholder alignment across researchers, managers, and project leads
- Built R and Python applications to predict and simulate manufacturing plant resource capacity across global sites, driving drug production planning and decision-making from requirements through deployment

Biostatistician

Nov 2021 – Jul 2023

L'Oréal R&I · Scientific Computing · via Astek / IT&M Stats

- Supported a senior statistician on exposome and clinical datasets, including biophysical, omics, and microbiome data
- Characterised clinical phenotypes with multi-block, integrative, multivariate, and variable-selection approaches
- Contributed to data-visualisation work, including Bayesian-network and other graphical approaches, while proposing suitable statistical methods
- Defined analysis methodologies for proof-of-concept clinical studies; wrote statistical analysis plans, supported blind-review meetings, programmed analyses in R, and delivered reports (100+ clinical studies analysed)

Earlier Experience

Data Scientist Intern — L'Oréal R&I · Augmented Beauty

Feb 2021 – Aug 2021

Modelled hair customer data for personalised recommendations, combining literature review, data management, statistical analyses, machine learning, and communication to Research&Innovation teams.

Biostatistics Project Officer Intern — Da Volterra · Metagenomics Team

Feb 2020 – Aug 2020

Performed data management and non-parametric statistical analyses for gut-microbiome clinical studies, then prepared and presented results to biotechnology research teams.

Data Scientist Intern — Flinders University · Digital Health Research Center

Sep 2019 – Jan 2020

Supported digital-health and wearables studies in R through questionnaire design, workshop preparation, statistical evaluation, and presentation of results to researchers.

Education

Diplôme d'Ingénieur — Statistiques appliquées aux Sciences de la Vie (2021)

Institut Agro - Agrocampus Ouest, Rennes

Master — Mathématiques Appliquées · Statistiques (2021)

INSA Rennes · Agrocampus Ouest · Université Rennes 2 · ENSAI · ENSAE (joint programme)

Technical Skills

Languages	Python · R · SQL · Bash
Deep learning & LLM	PyTorch · Hugging Face (transformers, PEFT, datasets) · LoRA / QLoRA · Instructor · faster-whisper · sentence-transformers · Ollama
ML & Stats	scikit-learn · tidymodels · XGBoost · Bayesian optimisation · DoE · PLS · PERMANOVA · survival analysis
Apps & Viz	R Shiny (golem) · FastAPI · ggplot2 · Plotly · Quarto · Streamlit
MLOps	Docker · GitHub Actions · Azure ML · Databricks · Posit Connect · renv · uv
Engineering	Git · CI/CD · automated testing · documentation · code review · architecture design
Domains	GxP · IQ/OQ/PQ · Clinical biostatistics · Multi-omics · Cosmetic R&D

Additional Information

Certifications

GitHub Foundations — GitHub (2024)

Scrum Master — Agilbee (2023)

Languages

French (native)

English (fluent, professional)